FPT UNIVERSITY • FPTU QUY NHON AI CAMPUS — FU CAHATA

FINAL CAPSTONE DEFENSE & PROJECT PRESENTATION (FALL 2026)

OMNICAST

Broadcast. Connect. Discover.

Enterprise media, real-time LiveStream and broadcast intelligence

PART 1: PRN232 | Back-End, gRPC & Next.js 15 (33 Rc

PART 2: PRM393 | Flutter Mobile App (31 Screens & Live Studio)

✓ 142,897 Physical LOC

✓ 468/468 Tests Passed in CI Run 40630318005

✓ WebRTC WHIP/WHEP + Socket.IO /live

Mentor / Supervisor: **Thầy Huỳnh Vang (Mr. Huỳnh Vang)**

Nguyễn Bình An • **Trần Thanh Sơn** (QE170080) • **Nguyễn Vĩnh Phú** (QE180136) • **Đồng Hoàng Nguyên**

FPTU Quy Nhon AI Campus — FU Cahata • October 2026

Project Team & Academic Mentorship

— FPTU Quy Nhon AI Campus — FU Cahata ● Mentor: Thầy Huỳnh Vang

1. NGUYỄN BÌNH AN (TomOutfit) — TEAM LEADER

- **Core Back-End:** NestJS 10 modular architecture (19 modules, 140+ REST endpoints).
- **Security & EPG Engine:** JWT + 4-role RBAC, atomic SQL schedule conflict detection, 24/7 seeded EPG filler, Direct Stream Engine & CI/CD.

2. TRẦN THANH SƠN (QE170080) — MICROSERVICES & DB

- **gRPC Microservice:** Standalone `audit-logger-service` over HTTP/2 Protocol Buffers (`audit_logger.proto`).
- **Database Engineering:** Supabase PostgreSQL (21 Prisma models), connection pooling (`:6543`), telemetry & seeding.

3. NGUYỄN VĨNH PHÚ (QE180136) — MOBILE FLUTTER LEAD

- **Flutter Client:** 4-Tier Clean Architecture, 12 BLoC/Cubits, 31 screens & 22 custom widgets.
- **Mobile Live & Offline:** `OmniPlayer`, `LanDiscoveryService`, `MobileStudioRecorderScreen` (WebRTC WHIP), SQLite offline sync.

4. ĐỒNG HOÀNG NGUYÊN (teng122) — WEB, LIVE & AI

- **Next.js 15 Portal:** 33 App Router screens, 7-theme `/live/[id]` lounge, 4-tab `/studio/live` console.
- **Realtime & AI:** Socket.IO `/live` gateway & OpenAI GPT-4o AI Curator with Zod validation + heuristic fallback.

Executive Summary — The Industry Challenge & The OmniCast Solution

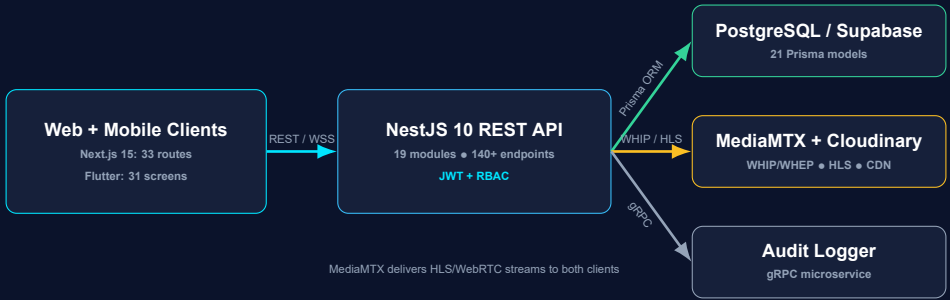
THE PROBLEM: BROADCAST COLLISION & FRAGMENTATION

- **Multi-Source Live Complexity:** Switching between VOD lineups, live studio cameras, and public streams often drops connected viewers.
- **Airtime Collisions & 24/7 Dead Air:** Overlapping events disrupt broadcasts, while sparse schedules leave blank EPG grids.
- **Manual Content Curation:** Broadcasters manually tag, rate, and check compliance across high-volume video catalogs.
- **Rigid Mobile Experiences:** Mobile players fail on local LAN IPs or freeze when offline.

THE OMNICAST SOLUTION: UNIFIED HYBRID ECOSYSTEM

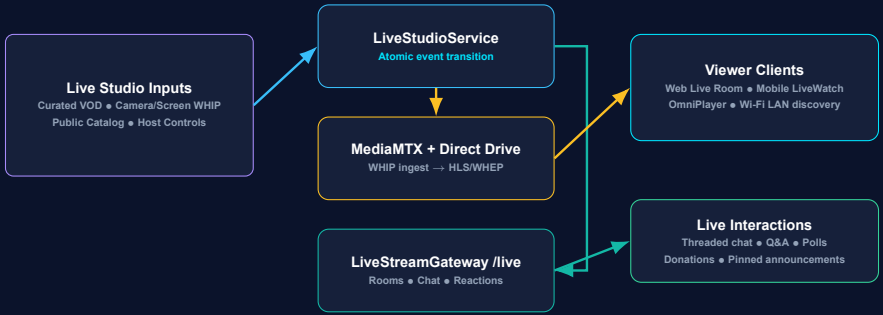
- **4-Mode Live Studio & Zero-Downtime Switch:** Curated VOD, WebRTC WHIP Camera/Screen, Public Catalog & instant program switching via Socket.IO `/live`.
- **Mathematical Collision & 24/7 EPG Engine:** Single raw SQL interval query + deterministic seeded 24/7 auto-filler.
- **AI Broadcast Curator:** Single-call GPT-4o curation with Zod schema validation and heuristic fallback.
- **LAN-Aware Offline-First Mobile App:** SQLite mutation queue + automatic Wi-Fi MediaMTX LAN discovery.

System Architecture — Web and Mobile Clients with Shared Backend Services



A unified monorepo architecture developed at **FPTU Quy Nhon AI Campus — FU Cahata** under Mentor **Mr. Huỳnh Vang**.

LiveStream & Live Studio Architecture — MediaMTX, WebRTC and Socket.IO



MODULE 1 OF 2 ● FPTU QUY NHON AI CAMPUS — FU CAHATA

PART 1: PRN232

Building Cross-Platform Back-End Application & Web Portal

NestJS 10 ● Next.js 15 (33 Routes) ● Live Studio & Live Room ● Supabase PostgreSQL ● gRPC ● AI Curator

Backend Architecture, Security & 5-Domain Database — NestJS 10 (19 Modules), 4-Role RBAC & 21 Prisma Models

19 MODULES & 5-DOMAIN DATABASE (21 MODELS)

- **1. Identity & Security:** `User`, `Role`, `RefreshToken`, `PasswordResetToken` (JWT 60m + 90d rotation).
- **2. Broadcast Core:** `LiveChannel`, `LiveEvent`, `Recording`, `ProductionTag`.
- **3. Social & Live Monetization:** `Comment`, `CommentLike`, `Reaction`, `Follow`, `Watchlist`, `WatchHistory`, `StreamDonation`.
- **4. Operations & gRPC Audit:** `AuditLog`, `Notification`, `DeviceToken`, `NewsletterSubscriber`.
- **5. AI Intelligence:** `AiCuratorReport` (1-to-1 with provenance).

SECURITY PIPELINE & RBAC MATRIX

- **Role-Based Access Control (4 Roles):**
  - `ADMIN`: System governance, user RBAC, gRPC audit log explorer.
  - `STAFF`: Live Studio 4-mode broadcasting, on-the-fly program switch, AI curation, schedule management.
  - `VIEWER`: Live room chat, floating reactions, 6-tier Super Chat donations, watchlist & history.
  - `GUEST`: Public EPG grid, channel catalog & VOD discovery.

Algorithmic Core: Collision Prevention & 24/7 EPG Filler — Raw SQL Effective-End

Interval Check & Seeded Deterministic Playout

1. ATOMIC SQL SCHEDULE CONFLICT DETECTION

Two broadcasts A and B collide iff $\max(\text{start}_A, \text{start}_B) < \min(\text{end}_A, \text{end}_B)$.

```
SELECT id, title FROM "LiveEvent"
WHERE "channelId" = $1::uuid
AND status IN ('SCHEDULED', 'LIVE')
AND "scheduledAt" < $newEnd
AND COALESCE("endedAt",
"scheduledAt" + ("duration" || ' min')::interval)
> $newStart;
```

● Handles `NULL endedAt` on active events and returns HTTP `409 Conflict` with colliding event details.

2. SEEDED 24/7 EPG AUTO-FILLER ENGINE

- **Zero Empty Days:** Guarantees full 24-hour (00:00–23:59) coverage for every channel and date.
- **Seeded PRNG:** Keyed on `(date, channelId)` so every client renders the exact same schedule.
- **Genre-Aware & Episode Splitting:** Prioritizes matching channel category, splits long VODs into episodes, and caps daily repeats (`MAX_REPEATS_PER_ITEM_PER_DAY`).
- **Pure Function:** Zero I/O side effects, verified by 18 unit tests.

AI Broadcast Curator & gRPC Audit Microservice — Single-Call GPT-4o + Zod Schema

Validation + Binary HTTP/2 gRPC Telemetry



CONSOLIDATED AI CURATOR (v4.0.0)

- Merges description, tags, compliance & suitability into 1 LLM call (`temp=0.3`, `top_p=0.95`), cutting latency by 66%.
- Records `promptVersion`, `aiModelVersion`, `openaiCallMs` and `openaiRetries`.

STANDALONE gRPC AUDIT MICROSERVICE

- Exposes `LogAudit`, `GetLogsByUser`, `GetLogsByAction`, `GetLogsByEntity`, `GetLogStats` over port `50051`.
- Non-blocking fire-and-forget RPC keeps main API latency unaffected.

Next.js 15 Web Portal: LiveStream Room & Live Studio — 33 Routes, 7

Genre-Themed Live Lounges & 4-Tab Broadcast Control Console

INTERACTIVE LIVESTREAM ROOM (`/live` & `/live/[id]`)

- **Live Hub (`/live`):** Auto-refreshes every 15s, displays total LIVE events & aggregate CCU pills, trending/newest sort.
- **7 Genre-Themed Lounges:** Dynamically styles room into *Kids*, *Podcast*, *Music Stage*, *Academy*, *Sports Arena*, *Obsidian Theater* or *Broadcast Lounge*.
- **Viewer Operations:** `JoinLobbyCard` entry, **Cinema Mode** (lights-off focus), multi-item **lineup program switcher**, real-time Socket.IO chat, **4 floating reactions** (♡, Fire, Clap, Wow) & pinned host banner.

4-TAB WEB LIVE STUDIO CONSOLE (`/studio/live`)

- **Tab 1 (Curated Live):** Pick 1 or more VOD recordings (`lineup`) with Anti-Piracy Guard & atomic transition.
- **Tab 2 (Camera / Screen / Upload):** `WebcamLiveBroadcaster` (WebRTC WHIP to MediaMTX), Screen Share (`getDisplayMedia`), or `UploadDropzone` (Signed Direct-to-Cloudinary upload with speed/ETA & canvas thumbnail).
- **Tab 3 (Verified Public Catalog):** 1-click import of verified HLS/MP4 streams with server `HEAD` probe.
- **Tab 4 (Host Console):** Broadcast pinned announcements (`broadcaster:message`) & stop live (`keepAsVod`).

MODULE 2 OF 2 • FPTU QUY NHON AI CAMPUS — FU CAHATA

PART 2: PRM393

Mobile Applications Development (Flutter 3.x & Dart)

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31 Screens • 12 BLoCs • OmniPlayer + LAN Discovery • Mobile Live Studio & Recorder • Offline SQLite

Mobile Clean Architecture & 31-Screen Hierarchy — 4-Tier Clean Architecture, 12

BLoC/Cubits & Declarative GoRouter Topology

4-TIER CLEAN ARCHITECTURE & 12 BLOCS

- **Presentation (31 Screens, 22 Widgets):** 6-tab `MainNavigationShell` (`Home`, `Channels`, `Schedule`, `Saved`, `History`, `Profile`) + Auth, Live, Studio & Admin clusters.
- **Logic (12 BLoC/Cubits):** `AuthBloc`, `ChannelBloc`, `EpgBloc`, `WatchlistBloc`, `LiveEventsCubit`, `ThemeCubit`, `LanguageCubit` with `droppable`/`restartable` transformers.
- **Data & Core:** 10 Repositories, Dio `AuthInterceptor` (401 queue lock & refresh), SQLite `omnicast_local.db`.

OFFLINE-FIRST SQLITE & LOCAL ALARMS

- **Offline Watchlist & EPG Cache:** Reads seamlessly from local SQLite when network drops.
- **Pending Mutation Queue (`pending_ops`):** Queues offline add/remove actions and replays FIFO upon reconnection.
- **15-Minute Advance Reminders:** Schedules exact local OS notifications 15 minutes before a bookmarked show airs.

Mobile LiveStream Experience: `LiveWatchScreen` & `OmniPlayer` — Wi-Fi LAN

Host Discovery, Zero-Cartoon Retry & 6 Interactive Live Operations

MULTI-ENGINE `OmniPlayer` & LAN DISCOVERY

- **Automatic Wi-Fi LAN Discovery (`LanDiscoveryService`):** Scans the local `/24` subnet on port `8888` to auto-replace `localhost`/`10.0.2.2` with the real MediaMTX host IP on physical phones.
- **Zero-Cartoon-Fallback Live Retry:** For live camera feeds (`/live/webcam_*`), performs up to 10 exponential-backoff retries without falling back to sample VODs.
- **Direct Drive & VIP Paywall:** Rewrites Google Drive URLs to `drive.usercontent.google.com` and enforces `SubscriptionPaymentSheet` on VIP streams.

`LiveWatchScreen` (`/live/:id`) OPERATIONS

- **1. Lobby & Live CCU:** `LiveJoinPromptBanner` + real-time `live:viewerCount` badge.
- **2. Threaded Live Chat & Moderation:** Nested replies (`parentId`), comment likes & instant `STAFF`/`ADMIN` deletion (`chat:delete`).
- **3. 0ms Haptic Floating Reactions:** `HapticFeedback.lightImpact()` + optimistic delta + rising particle overlay (♡, Fire, Clap, Wow/Party).
- **4. 6-Tier Super Chat (`LiveDonationModal`):** Preset gift tiers, custom message & celebratory banner.
- **5. Pinned Host Toast & Instant Program Switch:** Handles `broadcaster:message` & `live:program:changed`.

Mobile Live Studio & Field Reporter Camera Viewfinder — `LiveStudioScreen` (4 Tabs) & `MobileStudioRecorderScreen` (WebRTC WHIP)

`LiveStudioScreen` (4 BROADCAST TABS)

- **Tab 1 — Curated Live:** Select channel, title/description, and pick 1 or more VOD recordings (`lineup`) to go live immediately.
- **Tab 2 — Self-Record & Upload:** Launch `MobileStudioRecorderScreen` or upload device video to Cloudinary via signed parameters (`IMMEDIATE`, `SCHEDULED`, `PRIVATE`).
- **Tab 3 — Verified Public Catalog:** Browse & 1-click import verified HLS/MP4 streams or custom URLs.
- **Tab 4 — Host Console:** Send pinned announcements (`broadcaster:message`) & stop active broadcast (`keepAsVod`).

`MobileStudioRecorderScreen` VIEWFINDER

- **Dual Broadcast Modes:**
  - `Push-to-Live`: Real-time WebRTC WHIP streaming via `WebRTCPublisherService` with flashing red `ON-AIR` tally border.
  - `Record-then-Live`: 1080p MP4 capture + post-record **Master QC Review** player.
- **Professional Broadcast HUD:** Real-time **SMPTE Timecode** (`HH:MM:SS:FF`), dual-channel **Stereo VU Audio Meter**, **Rule-of-Thirds** & **Safe Area** guides, **Lower Third** overlay, `1x--5x` zoom, torch & auto-scrolling **Reporter Teleprompter HUD**.

Cross-Platform Synergy & Verification Matrix — 142,897 Physical LOC ● 468/468 Tests

Passed in Archived CI Run

Dimension	PRN232 (Back-End, gRPC & Web)	PRM393 (Flutter Mobile Client)
Core Stack	NestJS 10 + gRPC + Next.js 15 (33 Routes)	Flutter 3.x + Dart 3 (31 Screens, 22 Widgets)
Architecture	19 Domain Modules + Standalone gRPC Service	4-Tier Clean Architecture + 12 BLoC/Cubits
Database / Cache	Supabase PostgreSQL (21 Prisma models)	SQLite (`omnicast_local.db` + `pending_ops`)
LiveStream & Studio	7-Theme Live Room + 4-Tab Web Studio (WHIP)	`LiveWatchScreen` + 4-Tab Studio + ENG Recorder
Realtime Operations	Socket.IO `/live`: Chat, Emoji, Super Chat, Switch	0ms Haptic Emoji, 6-Tier Super Chat, LAN Scan
Code Volume	**79,568 Physical LOC**	**53,922 Physical LOC** (+ 9,407 Ops/Docs)

Live Deployment & Evaluation — Production Cloud Infrastructure & Pre-Seeded Test Credentials

LIVE PRODUCTION ENDPOINTS

- **Web Portal (Vercel Live):** `https://omnicast-fe.vercel.app`
- **Core API Swagger Documentation:** `https://omnicast-api.vercel.app/swagger`
- **Mobile APK Release (v1.2.0):** `github.com/TomOutfit/.../releases/tag/v1.2.0`
- **Official GitHub Repository:** `github.com/TomOutfit/OmniCast_PRN232-PRM393...`

PRE-CONFIGURED EVALUATION ACCOUNTS

Password for all test accounts: `Admin123!`

- **Admin Account (Full RBAC):** `admin@omnicast.tv`
- **Staff Account (Live Studio Broadcaster):** `staff@omnicast.tv`
- **Viewer Account (Live Chat & Super Chat):** `viewer1@omnicast.tv`

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Thank You for Your Attention!

Broadcast. Connect. Discover.

OmniCast: Enterprise Media & Broadcast Intelligence Network

Nguyễn Bình An • **Trần Thanh Sơn** (QE170080) • **Nguyễn Vĩnh Phú** (QE180136) • **Đồng Hoàng Nguyên**

We are ready for Live Demonstration & Committee Q&A

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